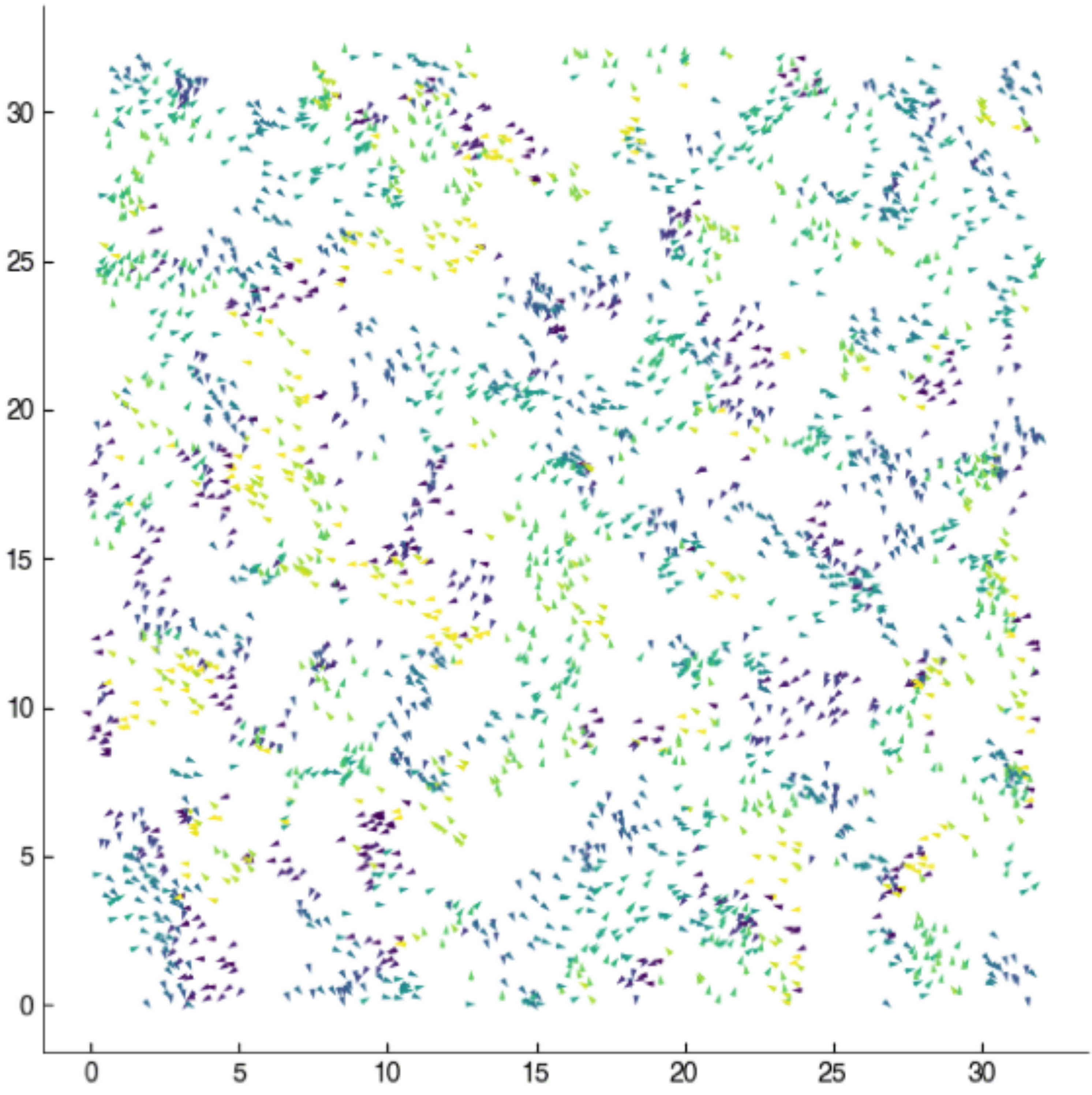
Active Vlasov interaction



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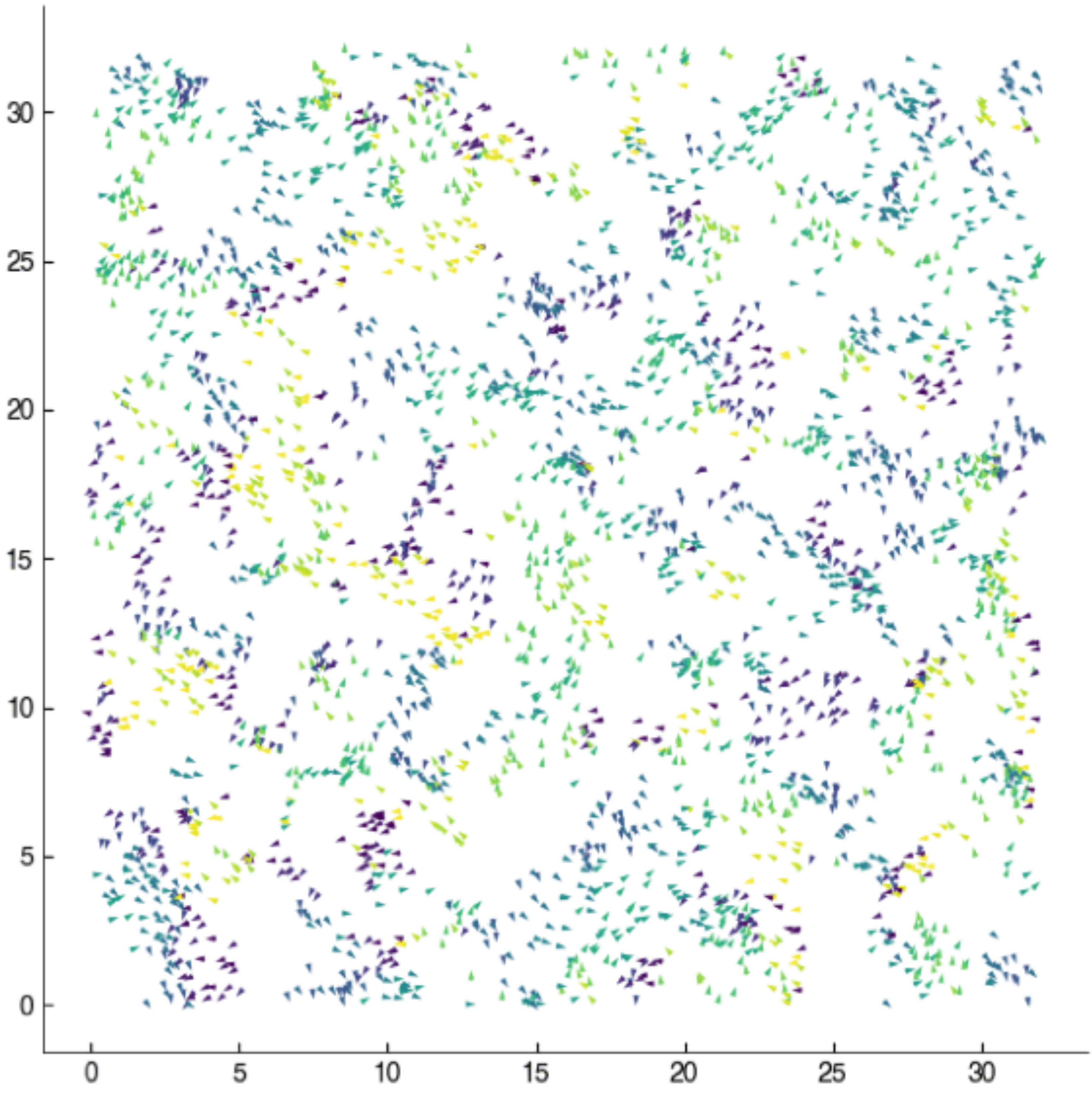
$$
\begin{cases}
\mathrm{d}X_t^i & = \mathrm{Pe}\, v(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\
\mathrm{d}\theta_t^i & = \chi F\left((X_t^i, \theta_t^i), \mu_t^N \right) \mathrm{d}t + \sqrt{2}\mathrm{d}B_t^i
\end{cases}
$$

- Interaction strength $\chi > 0$
- Empirical measure: $\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{(X_t^i, \theta_t^i)}$, McKean-Vlasov
- Famous interacting active Brownian particle model: Vicsek model (Phys. Rev. Lett., 1995) for flocking birds, 6000+ citations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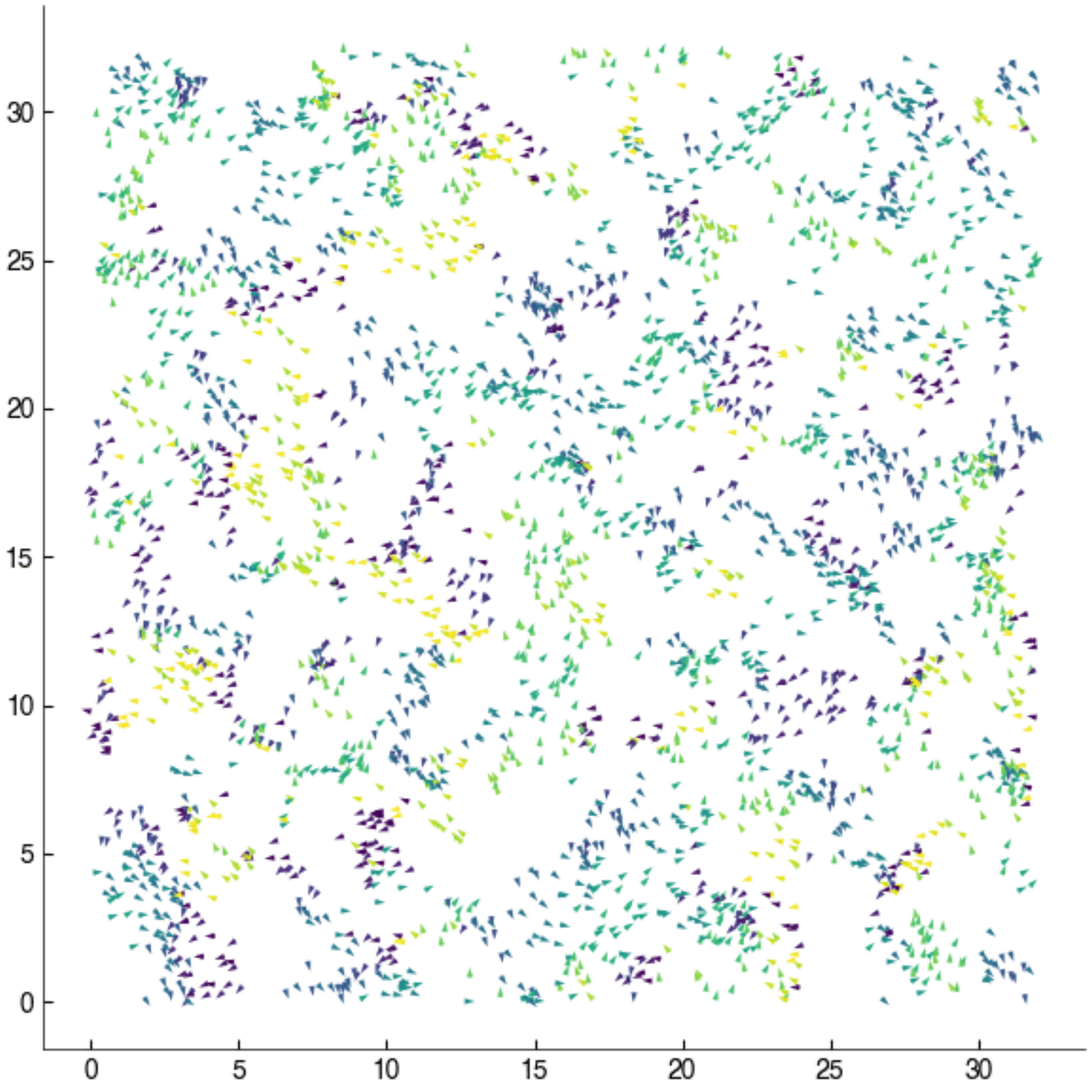

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Active Vlasov interaction

$$\begin{cases} \mathrm{d}X_t^i & = \mathrm{Pe}\,v(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i & = \chi F\left(\left(X_t^i, \theta_t^i\right), \mu_t^N\right)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

- Interaction strength $\chi > 0$
- Empirical measure: $\mu_t^N = \frac{1}{N}\sum_{i=1}^N \delta_{(X_t^i,\theta_t^i)}$, McKean-Vlasov
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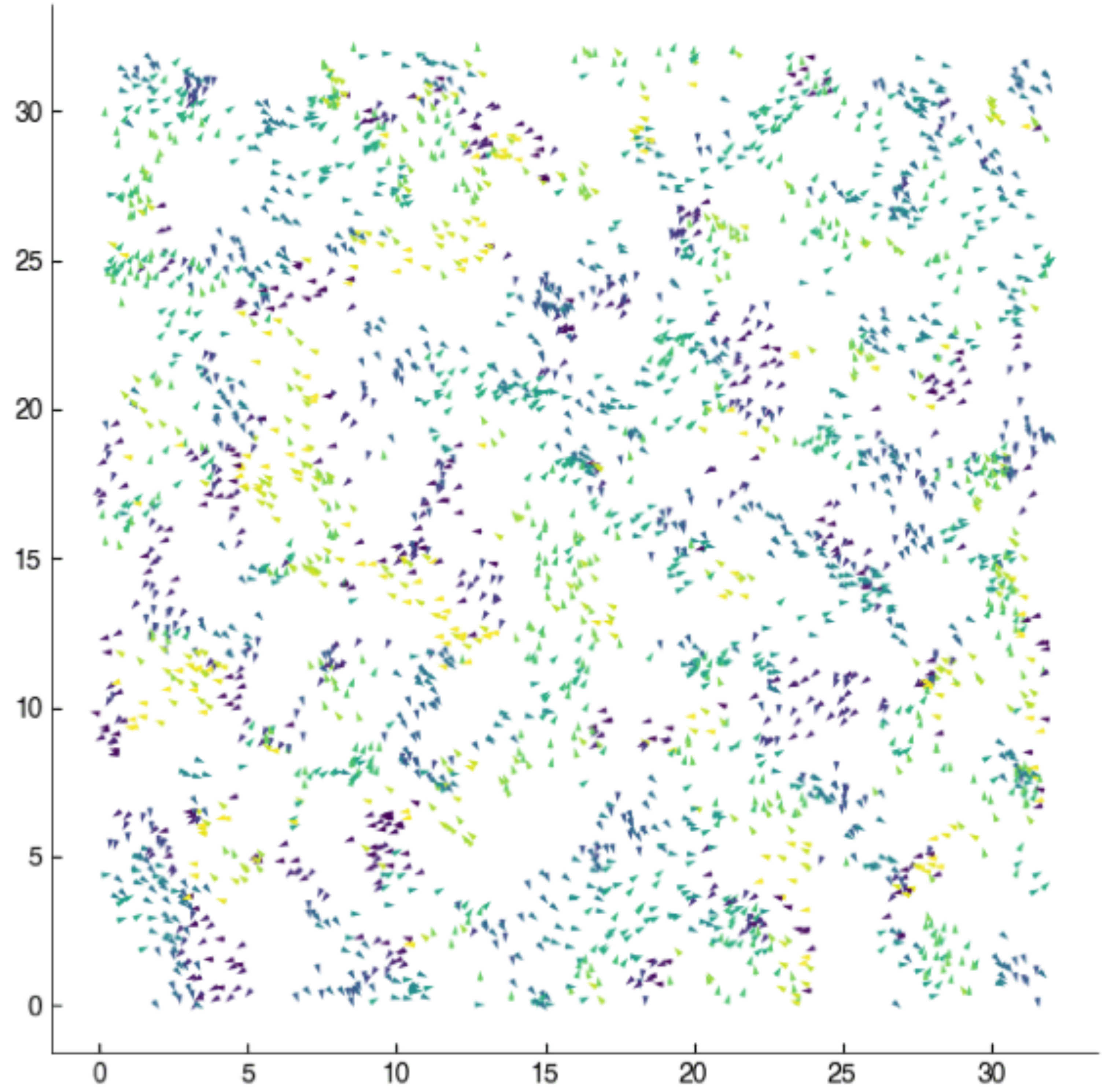

Vicsek model simulation © F. Turci 2020

Active ants

$$\begin{cases} \mathrm{d}X_t^i &= \mathrm{Pe}\, v(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \chi F\left(\left(X_t^i, \theta_t^i\right), \mu_t^N\right) \mathrm{d}t + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

- Ant model (dW, Bruna & Burger, SIAM J. Appl. Dyn. Syst., 2025):
  $F = n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j)$